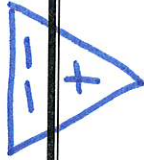


Dividend = The number being divided.

Divisor = The number of groups, or the size of the group.

Groups CANNOT be Negative!

Equation 1	Model	Equation 2
$-15 \div (-3) = 5$ ↓ Total ↓ size ↓ (Dividend) (Divisor) (Quotient)		$-15 \div 5 = -3$ ↓ Total ↓ groups ↓ size ↓ group
$-14 \div (-7) = 2$ ↓ Total ↓ size ↓ groups		$-14 \div 2 = -7$ ↓ Total ↓ groups ↓ size
$-12 \div (-3) = 4$ ↓ Total ↓ size ↓ groups		$-12 \div 4 = -3$ ↓ Total ↓ groups ↓ size
$-28 \div (-7) = 4$ ↓ Total ↓ size ↓ groups		$-28 \div 4 = -7$ ↓ Total ↓ groups ↓ size



Division Rules:

$$- \div - =$$

Based on the Equation 1 column, when the dividend and divisor have the **same sign**, the quotient is positive

$$- \div + =$$

Based on the Equation 2 column, when the dividend and divisor have **different signs**, the quotient is negative

1. $42 \div (-7) = -6$

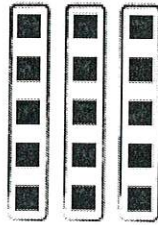
2. $\frac{-48}{12} = -4$

3. $\frac{-35}{-7} = 5$

4. $-64 \div (-8) = 8$

Practice: Write a multiplication or division expression to represent the problem and then solve.

1. Write a division and a multiplication expression to represent the model and solve.



$$-15 \div 3 = -5$$

$$3 \times -5 = -15$$

2. A computer stock lost 2 points each hour. Find the total change in stock after 6 hours.

$$6(-2) = -12$$

3. A deep-sea diver descends below the surface at a rate of 60 feet each minute. What is the depth of the diver after 10 minutes?

$$10(-60) = -600\text{ft}$$

4. Ms. Rockwell lost money on an investment at a rate of \$4 per day. What is the change in her investment after two weeks? $2 \times 7 = 14$

$$-4(14) = -\$56$$

5. During a 5 day work week, the stock market decreased by 55 points. What was the average daily change in the market?

$$\frac{-55}{5} = -11$$

6. Mr. Diaz decreased the speed of his car by 30 miles per hour over a period of 10 seconds. What was the average change in speed each second?

$$\frac{-30}{10} = -3$$

7. Last week, Ms. Cho found that the temperature outside dropped a total of 35 degrees. What was the average change in temperature each day?

$$\frac{-35}{7} = -5$$

8. Write a division and a multiplication expression to represent the model and solve.



$$-21 \div 7 = -3$$

$$7 \times -3 = -21$$